C#

C# Text File Handling

Mastering Reading and Writing for Beginners

Part 1: Writing to Files

Storing your data permanently in simple text files.

THE "ONE-LINER" METHODS

Fast & Efficient

Use the File class for quick tasks. These methods create the file automatically if it doesn't exist.

- **WriteAllText**: Overwrites everything with one string.
- **WriteAllLine**

```
// Example
string text = "Hello World"; File.WriteAllText("file.txt", text); string[] lines
```

APPENDING DATA

Don't want to lose existing data? Use AppendAllText.

This method adds new text to the end of the file without deleting what was already there. Ideal for logging events.

```
string log = "New login detected"; Fi
```

STREAMWRITER: FINE CONTROL

Professional Logging

For large files or continuous writing, StreamWriter is more efficient. It keeps the "stream" open until you're done.

- Better performance for large data.
- Support for

```
using (StreamWriter sw = new StreamWriter("p.txt")) { sw.WriteLine("First Line");
```

Part 2: Reading from Files

Retrieving your saved data back into the program.

READING MADE EASY

Reading Data

ReadAllText

Reads the entire file into a **single string**. Great for configuration files.

ReadAllLines

Reads every line into a **string array**. Perfect for processing lists line by line.

STREAMREADER: MEMORY EFFICIENT

Line by Line Processing

Reading a 1GB file into memory is bad! StreamReader reads one line at a time, keeping your app fast.

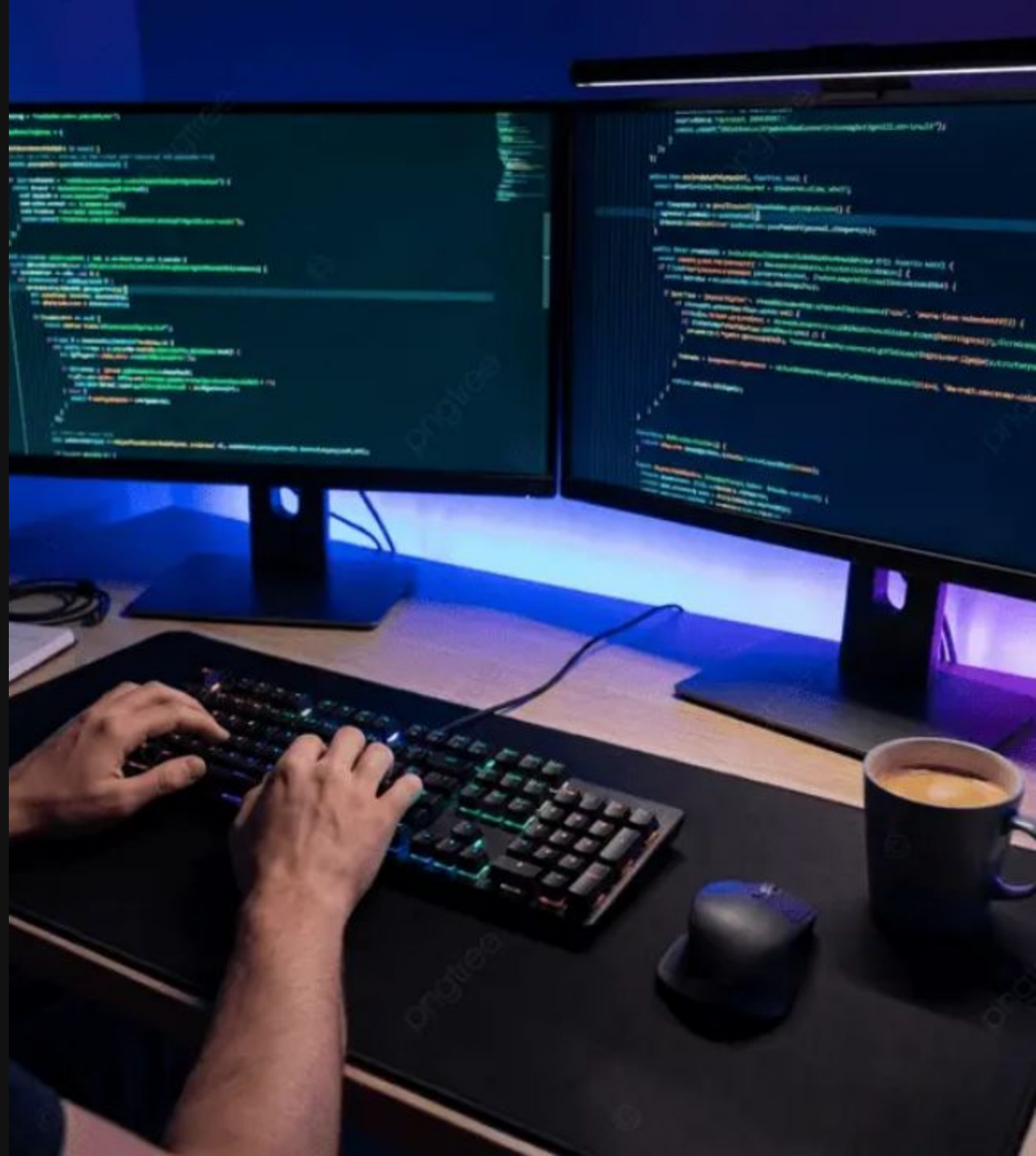
```
using (StreamReader sr = new StreamReader("big.txt")) { string line; while ((line = sr.ReadLine())
```

THE POWER OF 'USING'

When working with **Streams**, your app "locks" the file. If you forget to close it, other apps can't use it.

The using block automatically closes and disposes of the file as soon as the code inside is finished.

Always use it!



METHOD SELECTION GUIDE

Method	Type	Best For...	Overwrite?
WriteAllText	Write	Small strings / Single files	Yes
AppendAllText	Write	Logging / History	No
StreamWriter	Write	Large data / Loops	Optional
ReadAllLines	Read	Small lists (Arrays)	N/A
StreamReader	Read	Massive files (Line-by-line)	N/A

DON'T GET STUCK!

File Not Found

Always verify the path. Use File.Exists(path) before reading.

Access Denied

Check permissions. Files in "C:\" often require administrator rights.

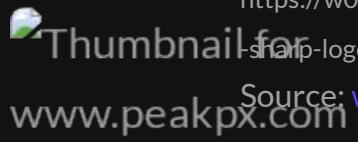
Memory Crash

Avoid ReadAllText on files over 100MB.

Happy Coding!

Any Questions on C# File Handling?

IMAGE SOURCES



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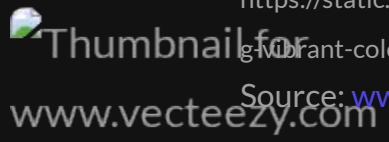

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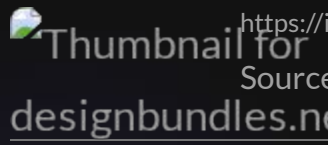

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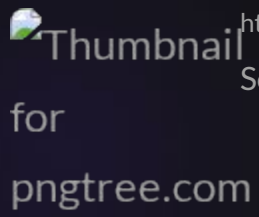

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